

Tutors



Open Web Learning Toolkit

Tutors Values



The ***Learner Experience*** prioritises web interactions that are **engaging, contextual, linkable, searchable, accessible** and **responsive**. In addition the experience should foster a sense of **community** and **connection** among fellow learners.

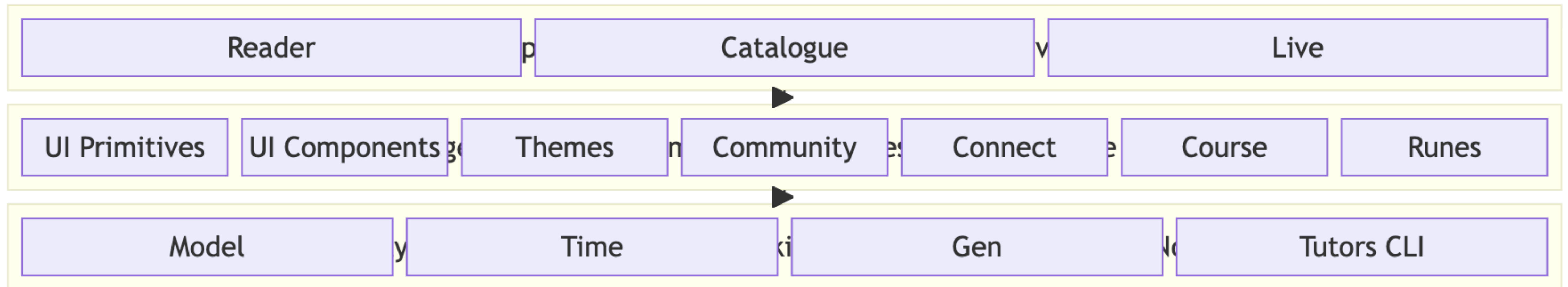


The ***Educator Experience*** prioritises the creation of a **guided paths** through a curriculum via the creation of learning materials that are **autonomous, structurally aligned, composable, auditable, extensible, versioned** and **independent**.

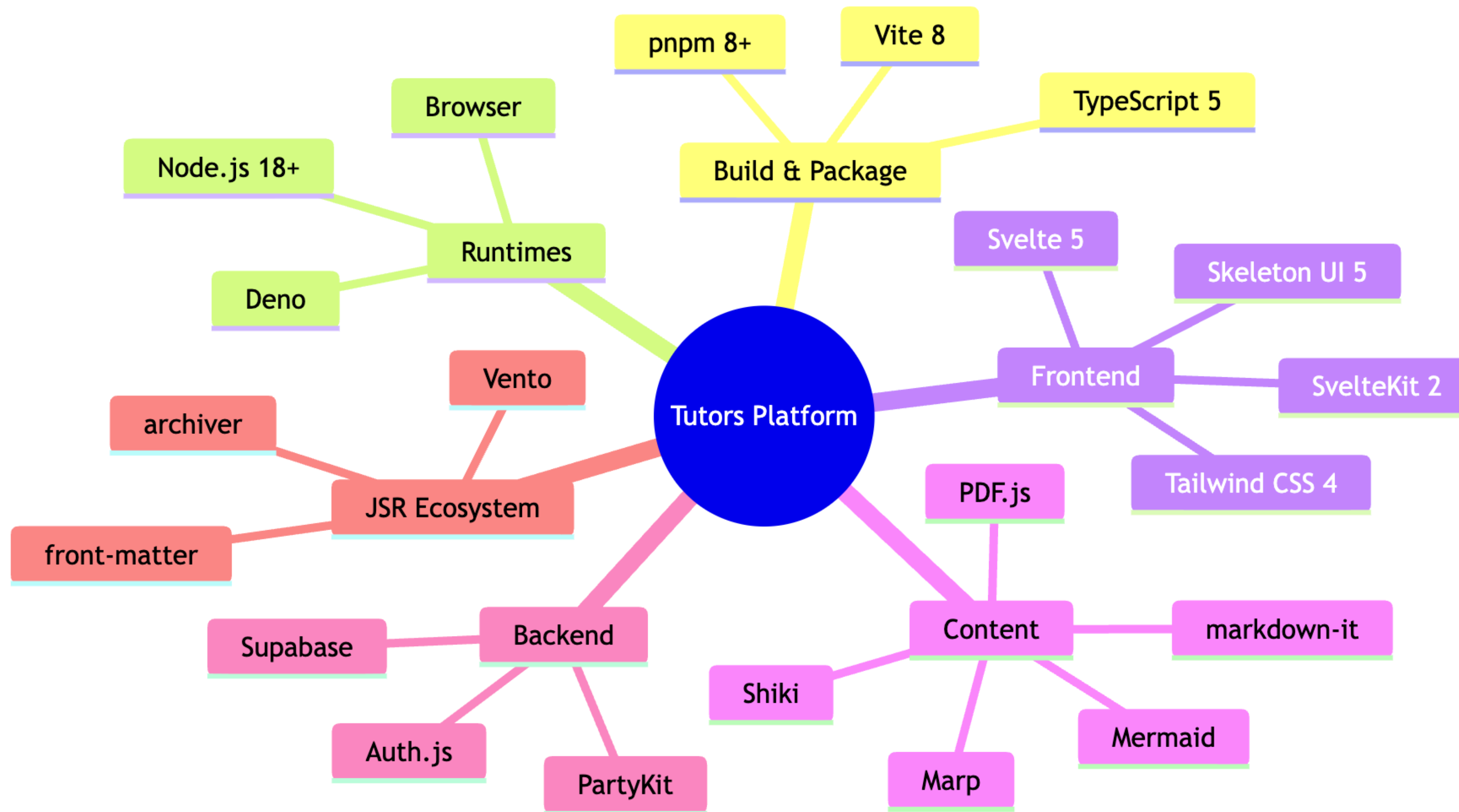


The ***Developer Experience*** prioritises the specification and implementation of **robust, well documented, loosely coupled components & services**, integrated into a **coherent toolkit** open to contributions from **diverse skill sets**.

Logical Structure



1. **Bottom-Up Dependencies:** Lower layers never depend on higher layers
2. **Package Isolation:** Each package has clear boundaries and responsibilities
3. **Shared Types:** JSR packages provide TypeScript types used across the ecosystem
4. **Runtime Compatibility:** JSR packages work in both Deno and Node.js
5. **Svelte-Specific:** Svelte packages leverage Svelte 5 reactivity



Build & Package Management:

- **pnpm 8+**: Fast, disk-efficient package manager with workspace support
- **Vite 8.x**: Next-generation build tool and dev server
- **TypeScript 5+**: Type-safe development across all packages

Runtime Environments:

- **Deno**: For JSR packages and CLI tools
- **Node.js 18+**: For Svelte packages and applications
- **Browser**: For client-side applications

Frontend Framework:

- **Svelte 5**: Reactive UI framework with runes-based state
- **SvelteKit 2**: Full-stack framework with SSR/CSR
- **Tailwind CSS 4**: Utility-first styling
- **Skeleton UI 5**: Svelte component library

Content Processing:

- **markdown-it**: Markdown parsing
- **Shiki**: Syntax highlighting
- **Marp**: Presentation rendering
- **Mermaid**: Diagram rendering
- **PDF.js**: PDF viewing

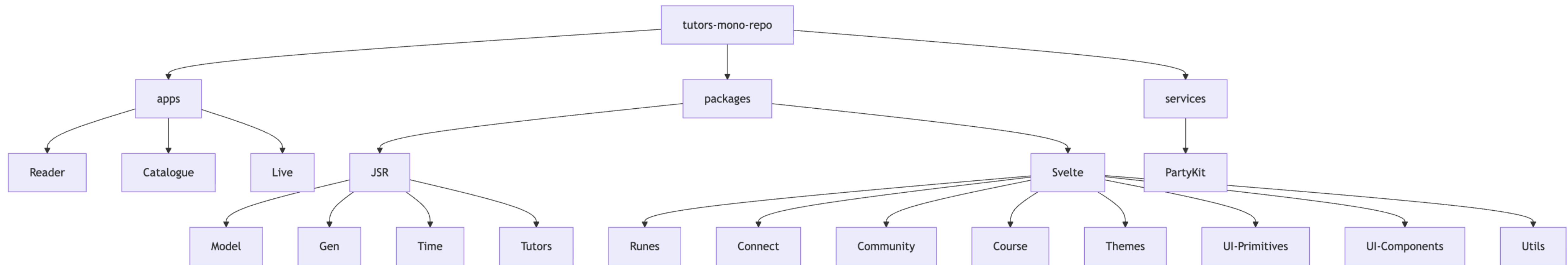
Backend Services:

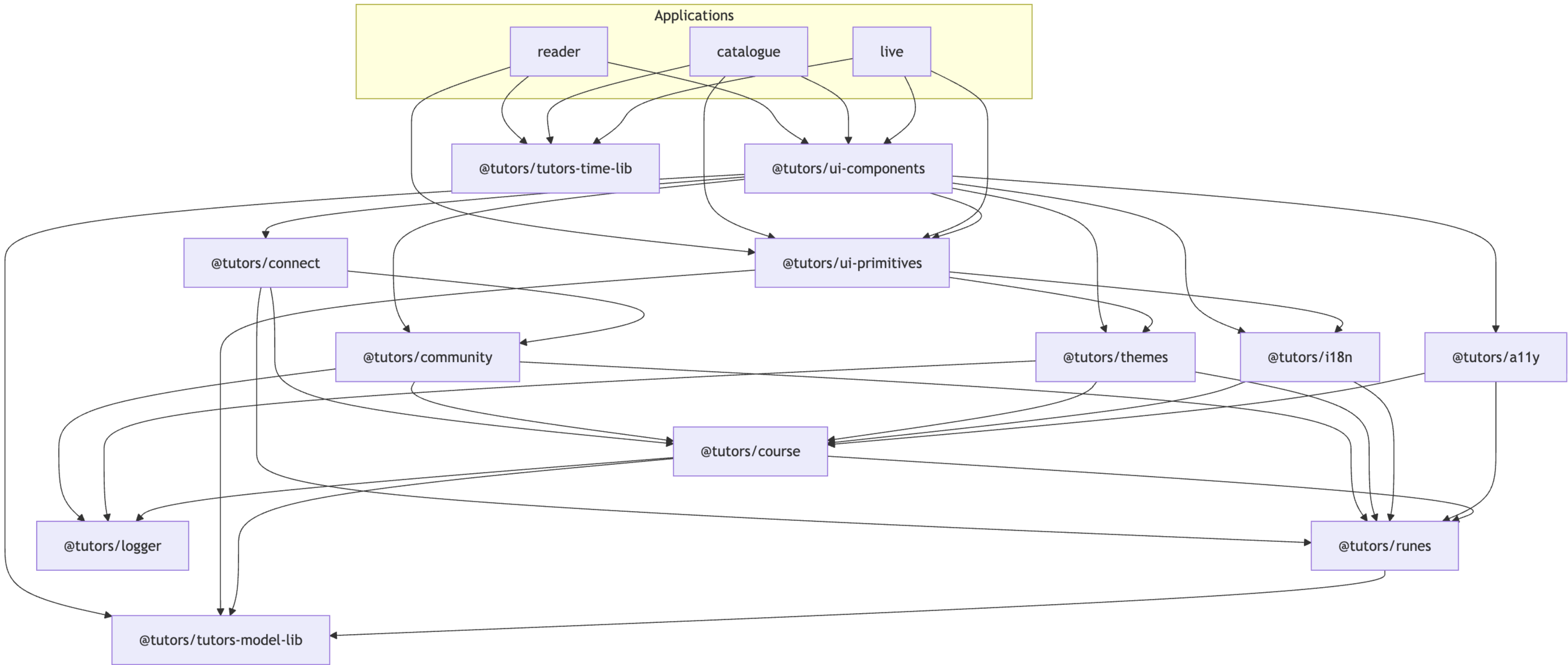
- **Supabase**: Database, authentication, analytics
- **PartyKit**: Real-time WebSocket communication
- **Auth.js**: GitHub OAuth integration

JSR Ecosystem (Deno-first):

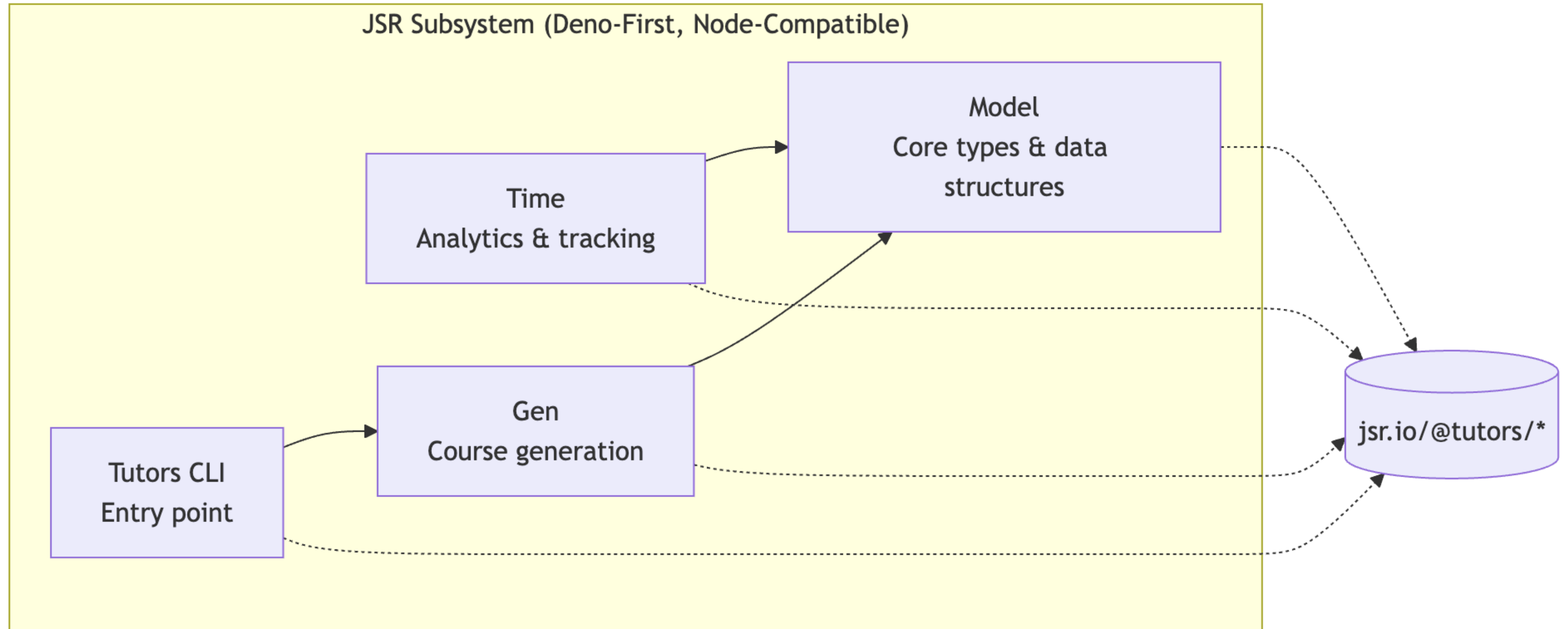
- **Vento**: Template engine for HTML generation
- **archiver**: ZIP file creation
- **front-matter**: YAML parsing

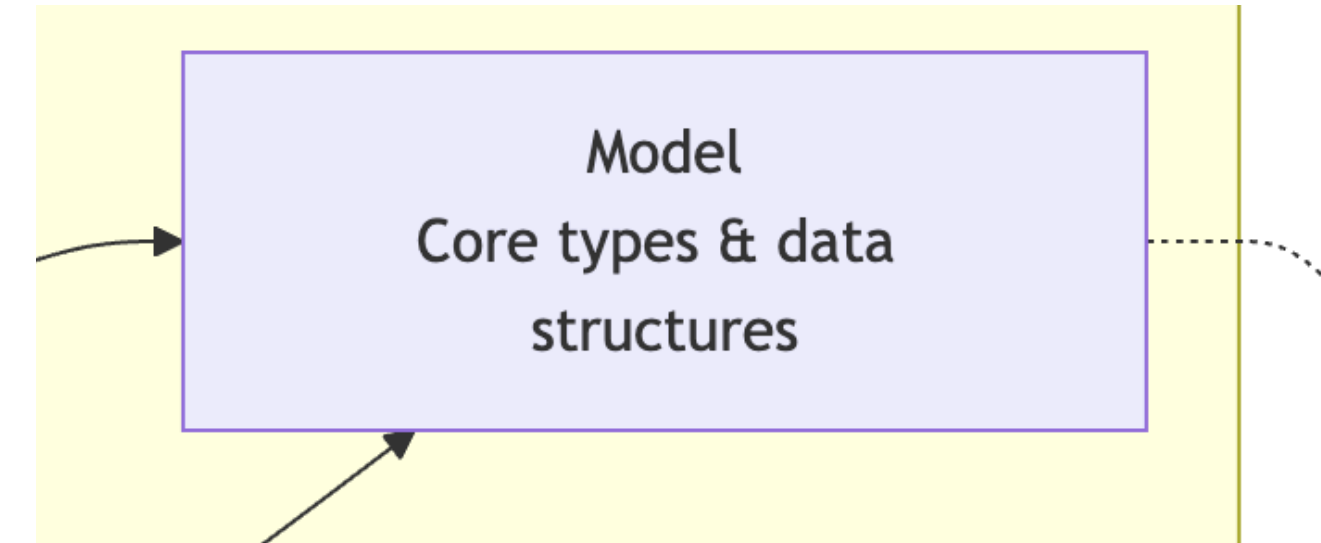
Tutors repo folder structure





Package Dependencies





1. Model Package (packages/jsr/model)

Package Name: @tutors/tutors-model-lib **Purpose:**

Core data structures, types, and utilities

Key Responsibilities

- **Type Definitions:** TypeScript interfaces for all learning objects
- **Data Models:** Course, Topic, Lab, Talk, Video, Note, etc.
- **Markdown Processing:** markdown-it configuration and plugins
- **Utility Functions:** Tree traversal, search, course utilities

```
packages/jsr/model/src/
├─ tutors.ts           # Main export
├─ types/
│   ├─ learning-objects.ts # Lo, Course, Topic types
│   ├─ icon-types.ts      # Icon configurations
│   ├─ media-types.ts     # Video, PDF types
│   ├─ calendar-types.ts  # Calendar structures
│   └─ tutors-id-types.ts  # Identifier types
├─ utils/
│   ├─ lo-utils.ts        # Learning object utilities
│   ├─ course-utils.ts    # Course tree manipulation
│   ├─ markdown-utils.ts  # Markdown processing
│   └─ markdown-plugins.ts # Custom markdown-it plugins
└─ services/
    ├─ lo-tree.ts         # Tree traversal and decoration
    └─ search.ts          # Search functionality
```


3. Gen Package (packages/jsr/gen)

Package Name: @tutors/tutors-gen-lib **Purpose:** Course content generation from markdown source

Key Responsibilities

- **Content Parsing:** Read markdown files and extract metadata
- **JSON Generation:** Create tutors.json course structure
- **HTML Generation:** Create static HTML for tutors-lite
- **Asset Processing:** Copy and organize course assets
- **Template Rendering:** Vento-based HTML templates

Key Components

CourseBuilder (services/course-builder.ts):

- Scans folder structure
- Identifies learning object types from folder/file patterns
- Builds hierarchical course tree

ResourceBuilder (services/resource-builder.ts):

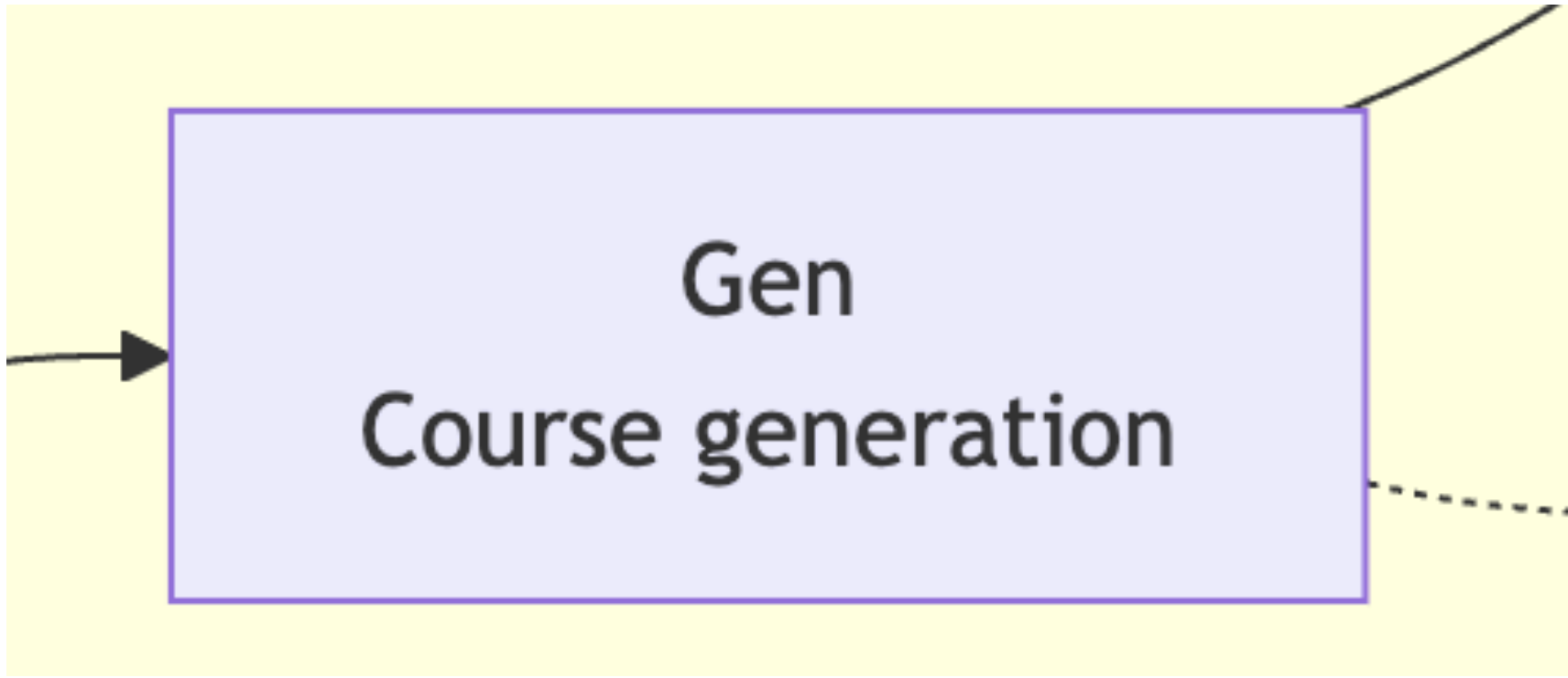
- Reads markdown frontmatter (YAML)
- Extracts title, summary, images
- Processes content markdown

TemplateEngine (templates/template-engine.ts):

- Loads Vento templates for HTML generation
- Renders Lo-specific views (Lab.vto, Talk.vto, etc.)
- Applies theme styles

CourseEmitter (services/course-emitter.ts):

- Writes tutors.json (for reader consumption)
- OR generates html/ folder (for tutors-lite)



```
packages/jsr/gen/src/
├─ services/
│   ├─ course-builder.ts    # Folder parsing
│   ├─ resource-builder.ts  # Metadata extraction
│   └─ course-emitter.ts    # Output generation
└─ templates/
    ├─ template-engine.ts   # Vento integration
    ├─ template-downloader.ts
    ├─ styles.ts             # CSS injection
    ├─ utils.ts
    └─ vento/                # HTML templates
        ├─ Composite.vto
        ├─ Lab.vto
        ├─ Talk.vto
        ├─ Topic.vto
        └─ components/
            ├─ cards/
            ├─ navigators/
            └─ ...
```

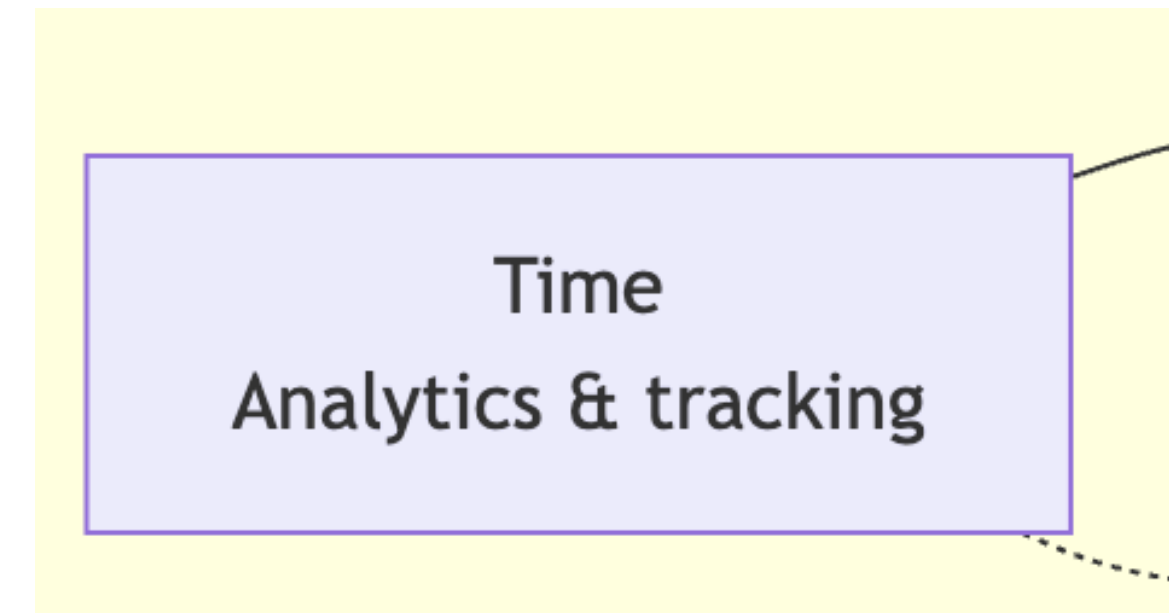
2. Time Package (packages/jsr/time)

Package Name: @tutors/tutors-time-

lib Purpose: Analytics, time tracking, and learning metrics

Key Responsibilities

- **Event Tracking:** Learning event capture and storage
- **Calendar Models:** Time-based activity views
- **Lab Analytics:** Step-by-step progress tracking
- **Supabase Integration:** Data persistence layer



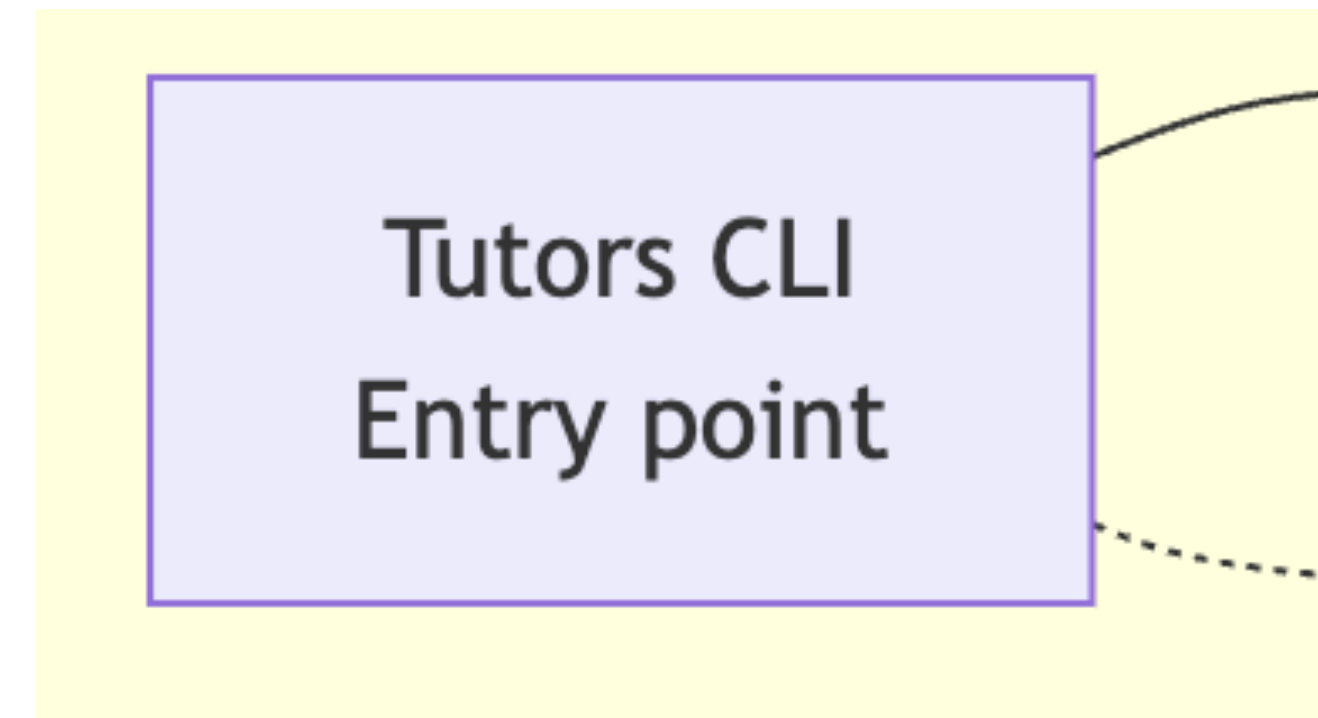
```
packages/jsr/time/src/  
├── index.ts           # Main export  
├── types/  
│   ├── tutors-time-types.ts  
│   ├── calendar-types.ts  
│   ├── lab-types.ts  
│   └── tutors-connect-types.ts  
├── services/  
│   ├── tutors-time.ts      # Main analytics service  
│   ├── course-time.ts     # Course metrics  
│   ├── base-calendar-model.ts  
│   ├── base-lab-model.ts  
│   └── supabase.ts        # Supabase client  
└── utils/  
    ├── calendar-utils.ts  
    └── lab-utils.ts
```

4. Tutors Package (`packages/jsr/tutors`)

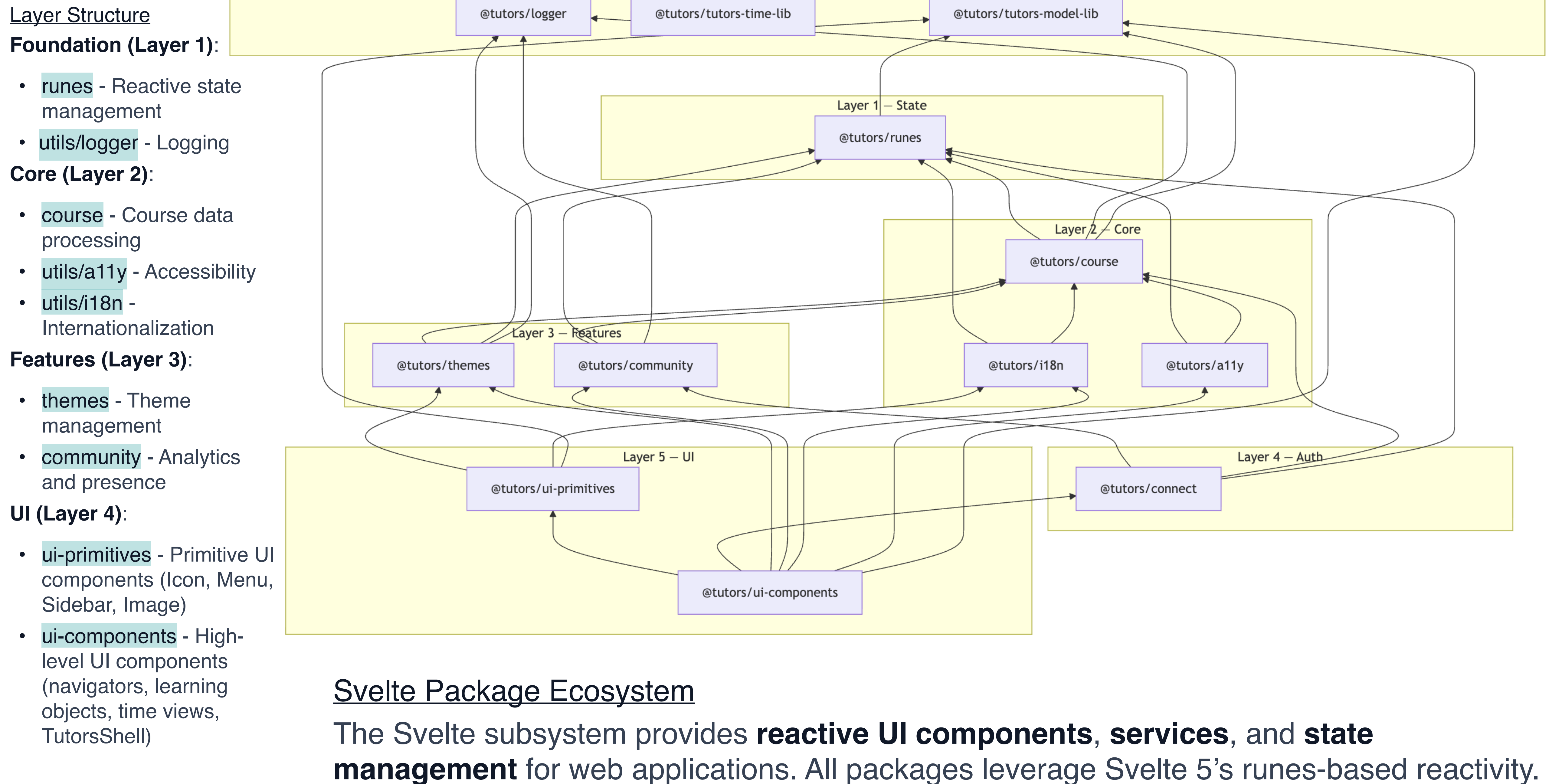
Package Name: `@tutors/reader` (or `@tutors/tutors`) **Purpose:** CLI entry point for course generation

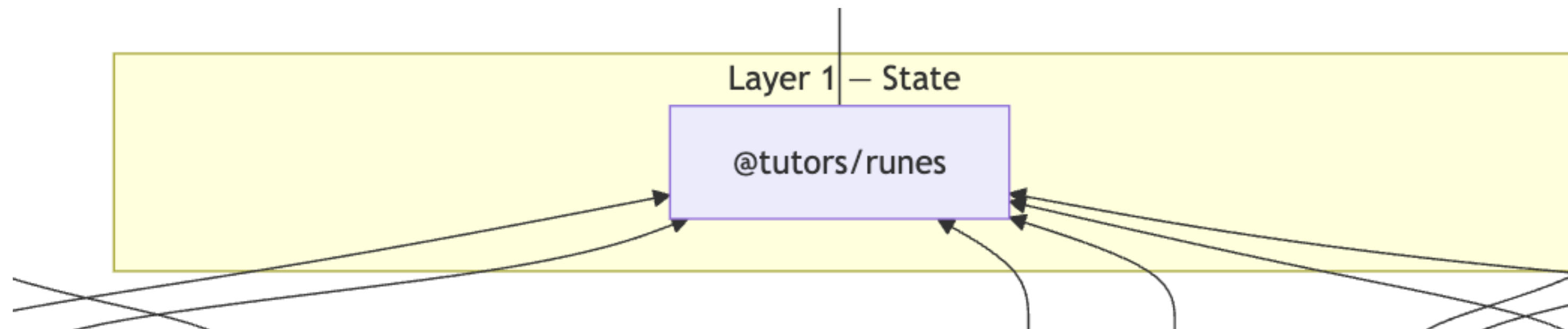
Key Responsibilities

- **CLI Interface:** Command-line tool for authors
- **Orchestration:** Coordinates gen package services
- **Configuration:** Reads Deno configuration and environment



```
packages/jsr/tutors/  
├─ deno.json           # JSR package configuration  
├─ main.ts             # CLI entry point  
└─ readme.md           # Usage documentation
```



```
// Lazy-initialized reactive state
export const currentCourse: {
  get value(): Course | null;
  set value(v: Course | null);
};

export const currentLo: {
  get value(): Lo | null;
  set value(v: Lo | null);
};

export const currentLabStepIndex: {
  get value(): number;
  set value(v: number);
};

export const layout: {
  get value(): "expanded" | "compacted";
  set value(v: "expanded" | "compacted");
};
```

1. Runes Package (packages/svelte/runes)

Package Name: @tutors/runes **Purpose:**
Global reactive state using Svelte 5 runes

2. Course Package (packages/svelte/course)

Package Name: @tutors/course **Purpose:** Course loading, processing, and markdown rendering

Sub-modules

Course Service (course/services/course.svelte.ts):

- Fetches course JSON from CDN
- Caches loaded courses
- Manages lab instances
- Decorates course tree with URLs

Live Lab (course/services/live-lab.ts):

- Manages lab step navigation
- Fetches markdown content for steps
- Converts markdown to HTML
- Tracks current step state

Live Notebook (course/services/live-notebook.ts):

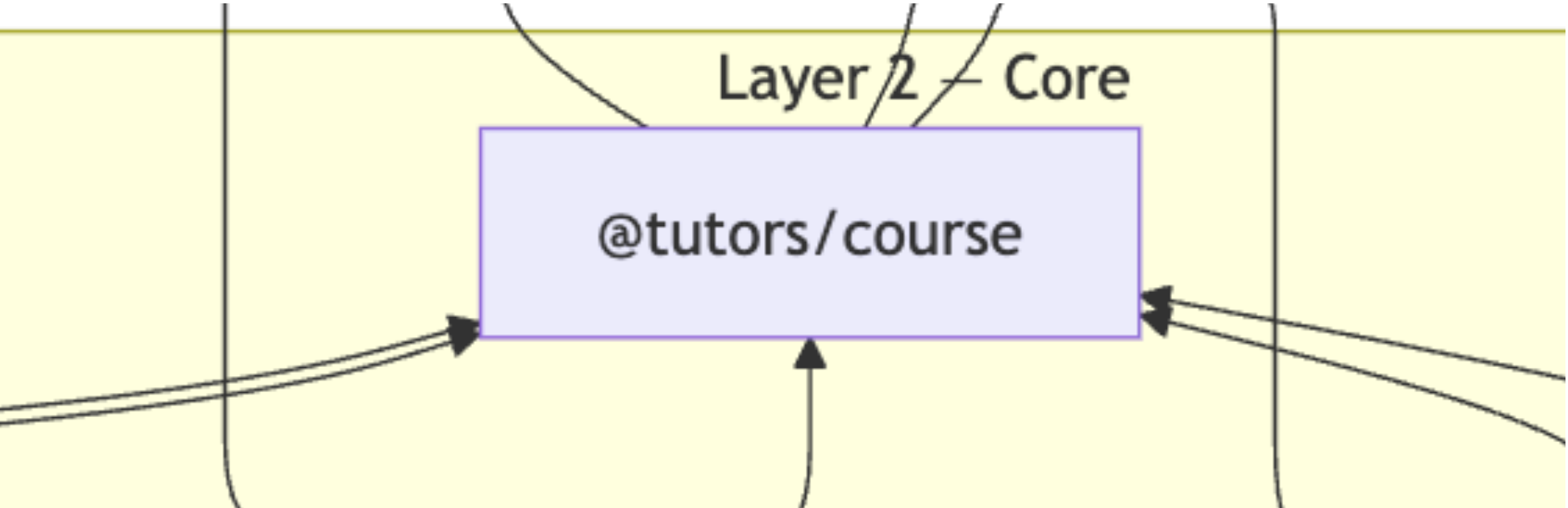
- Manages Jupyter notebook rendering
- Cell-by-cell content processing

Lo Tree (course/services/lo-tree.ts):

- Tree traversal utilities
- URL decoration
- Index building (topicIndex, labIndex, etc.)

Markdown Service (markdown/services/markdown.svelte.ts):

- Markdown-to-HTML conversion
- Syntax highlighting with Shiki
- Mermaid diagram rendering
- Marp presentation rendering
- Code block copy button action



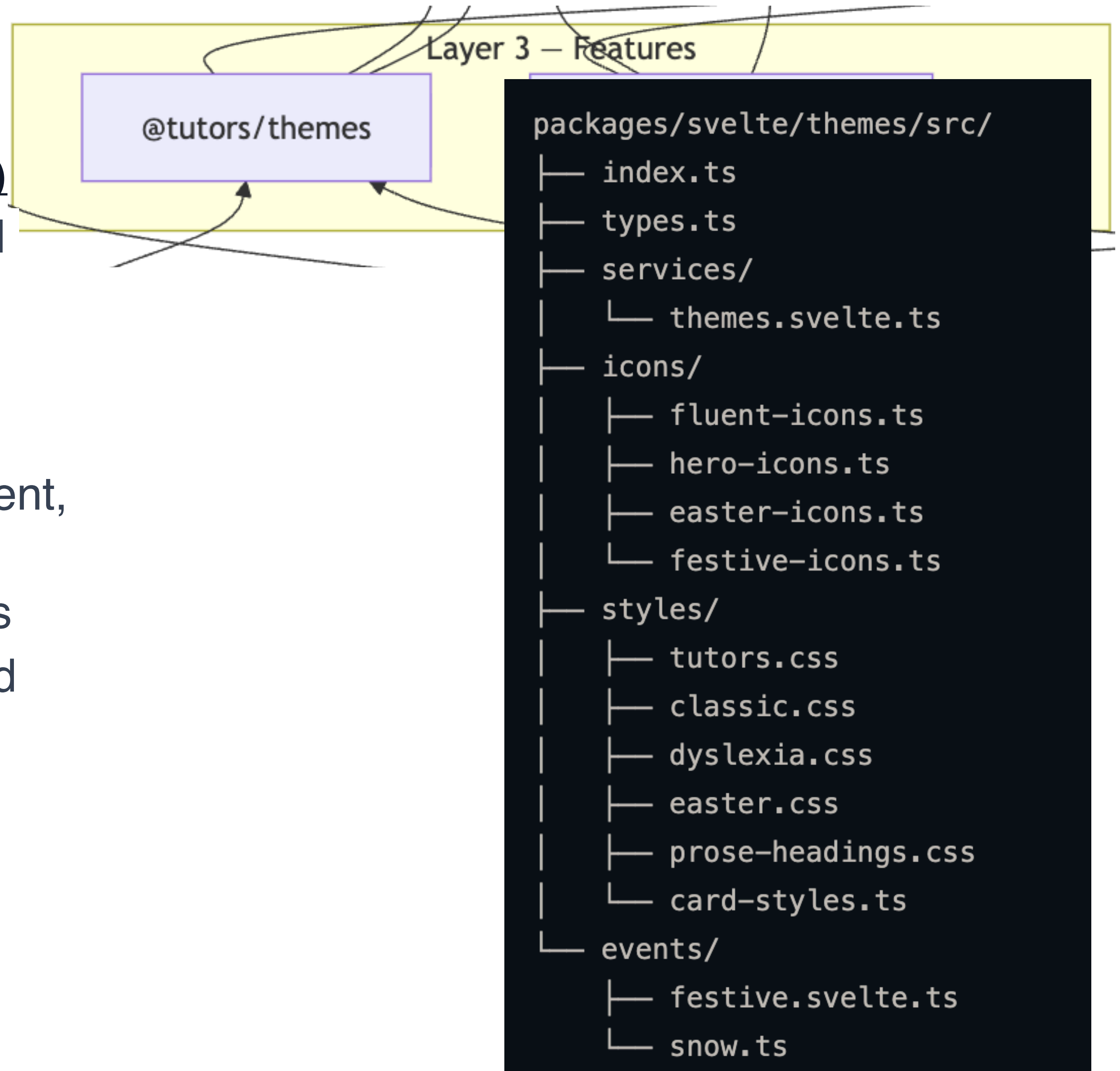
```
packages/svelte/course/src/
├── course/
│   ├── index.ts
│   ├── types.ts
│   └── services/
│       ├── course.svelte.ts
│       ├── live-lab.ts
│       ├── live-notebook.ts
│       └── lo-tree.ts
├── markdown/
│   ├── index.ts
│   ├── types.ts
│   ├── services/
│   │   ├── markdown.svelte.ts
│   │   ├── marp-renderer.ts
│   │   └── mermaid-action.ts
│   └── actions/
│       └── copy-code-action.ts
└── index.ts
```

3. Themes Package (`packages/svelte/themes`)

Package Name: `@tutors/themes` **Purpose:** UI theming, icons, and display modes

Features

- **Display Modes:** Light/dark theme toggle
- **Icon Sets:** Multiple icon theme choices (Fluent, Hero, Easter, Festive)
- **Layouts:** Expanded/compact card layouts
- **Card Styles:** Portrait/landscape/circular card modes
- **Festive Mode:** Snow animation effect

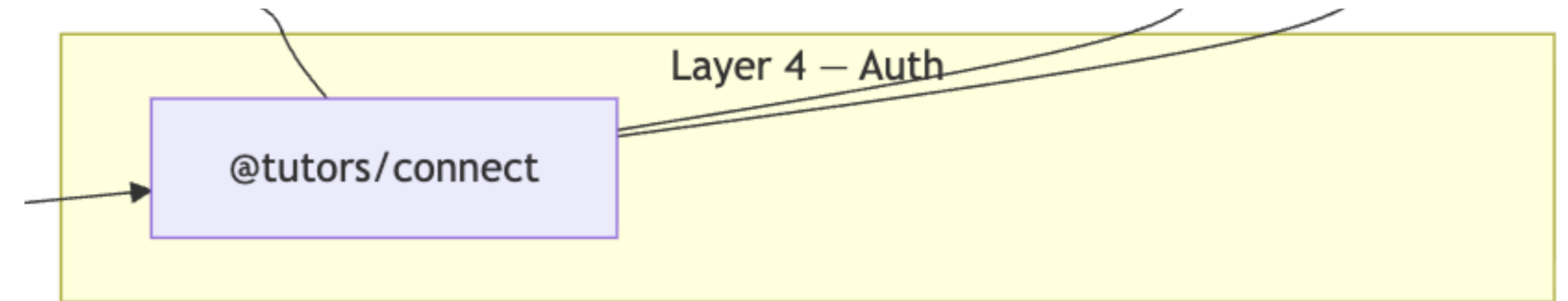


5. Connect Package (packages/svelte/connect)

Package Name: @tutors/connect **Purpose:**
User authentication and session management

Features

- GitHub OAuth integration
- User profile management
- Session persistence (localStorage or Supabase)
- Course access control



```
packages/svelte/connect/src/  
├─ index.ts  
├─ types.ts  
├─ services/  
│   └─ connect.svelte.ts  
│   └─ localStorageProfile.ts  
│   └─ supabaseProfile.svelte.ts  
└─ utils/  
    └─ allCourseAccess.ts
```


4. Community Package (packages/svelte/community)

Package Name: @tutors/community **Purpose:**
Analytics, real-time presence, and social features

Sub-services

Analytics Service:

- Tracks page views and learning events
- Sends data to Supabase

Presence Service:

- Course-specific student presence
- Real-time status updates via PartyKit

Live Service:

- Platform-wide live activity monitoring
- Online student count

Catalogue Service:

- Course directory management
- Visit statistics

```
packages/svelte/community/src/  
├─ index.ts  
├─ types.svelte.ts  
├─ services/  
│   ├─ analytics.svelte.ts  
│   ├─ presence.svelte.ts  
│   └─ live.svelte.ts  
└─ catalogue.ts  
└─ utils/  
    └─ supabase-client.ts
```

Layer 3 – Features

@tutors/community

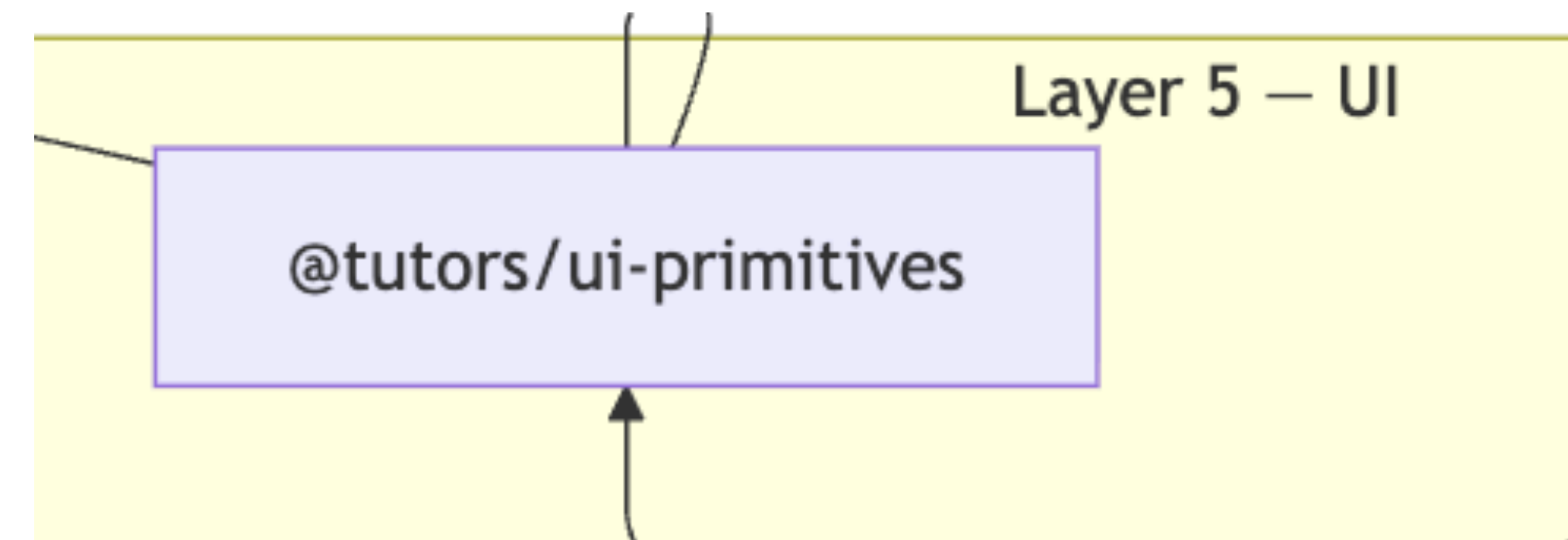
The diagram illustrates the relationship between the package structure and the Layer 3 - Features layer. A yellow box labeled 'Layer 3 - Features' contains a purple box labeled '@tutors/community'. Arrows point from the package structure (shown in a dark box) to the '@tutors/community' box, indicating that the package structure is the implementation of the features defined in the Layer 3 - Features layer.

6. UI Primitives Package (`packages/svelte/ui-primitives`)

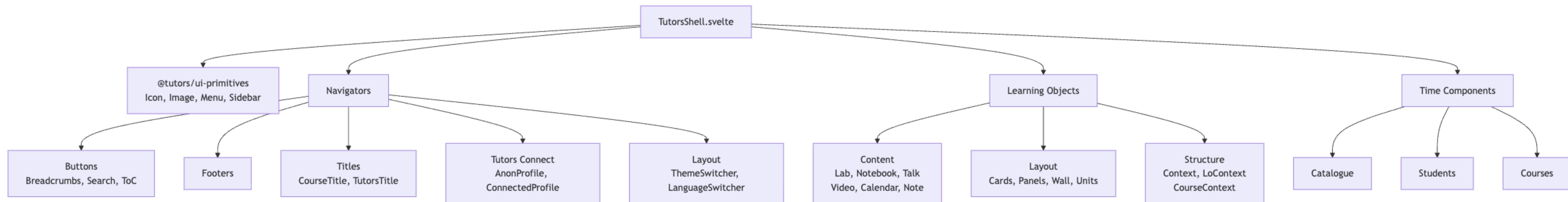
Package Name: `@tutors/ui-primitives` **Purpose:** Low-level, reusable primitive UI components

Components

- **Icon** - Iconify-based icon rendering with theme-aware type mapping
- **IconBar** - Horizontal icon strip
- **Image** - Responsive image with fallback
- **Menu / MenuItem** - Dropdown menu system
- **Sidebar** - Collapsible navigation sidebar
- **Setulcon / TutorsIcon** - Branding icons



```
packages/svelte/ui-primitives/src/  
├─ index.ts  
├─ _safelist.svelte  
└─ components/  
    ├─ Icon.svelte  
    ├─ IconBar.svelte  
    ├─ Image.svelte  
    ├─ Menu.svelte  
    ├─ MenuItem.svelte  
    ├─ Sidebar.svelte  
    ├─ SetuIcon.svelte  
    └─ TutorsIcon.svelte
```

7. UI Components Package ([packages/svelte/ui-components](#))

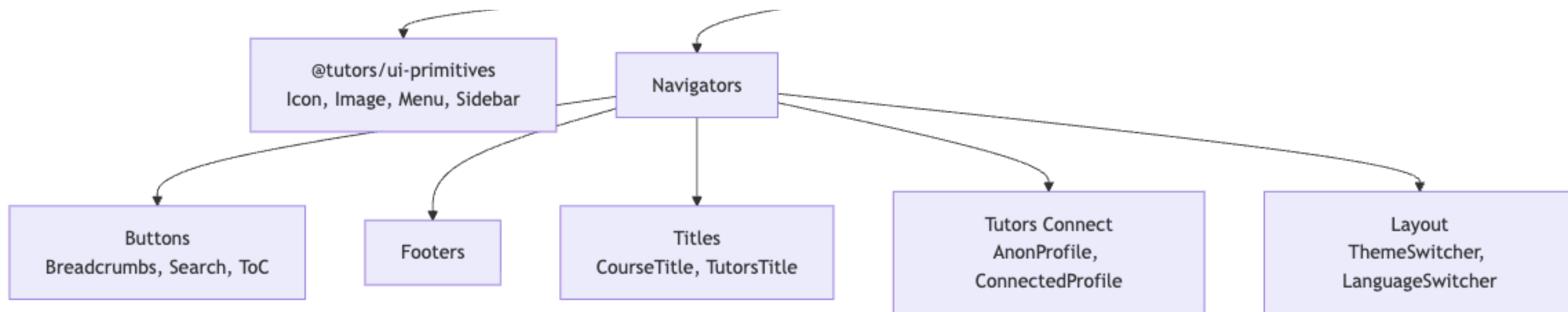
Package Name: [@tutors/ui-](#)

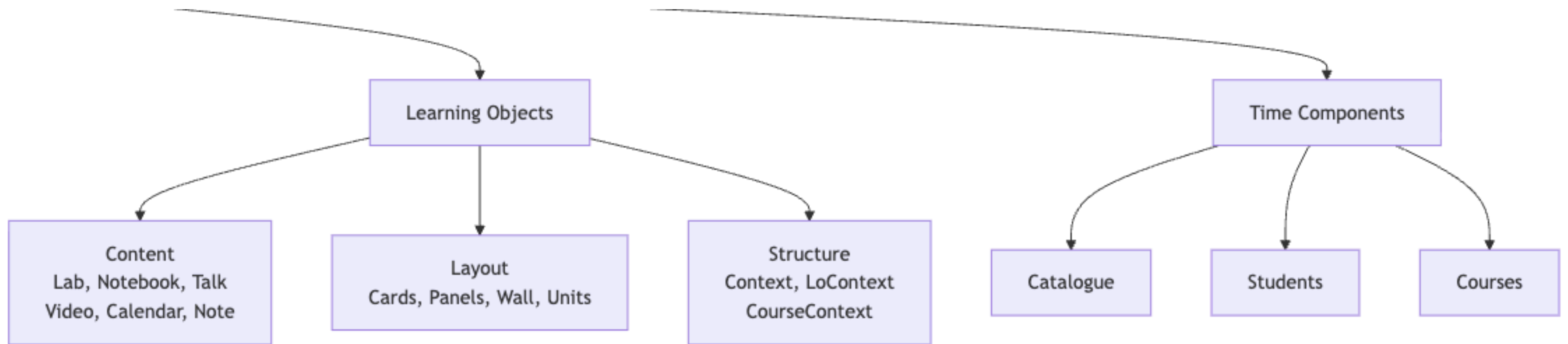
components Purpose: High-level, domain-specific UI components

This package depends on [@tutors/ui-primitives](#) and produces a pre-compiled CSS file ([dist/style.css](#)) containing all Tailwind utilities and Skeleton theme styles required by the applications. It must be built before running any app.

```

packages/svelte/ui-components/src/
├── TutorsShell.svelte      # Main app shell
├── styles.css              # Tailwind/Skeleton entry point
├── _dynamic-classes.css   # Safelist for @apply directives
├── _safelist.svelte       # Safelist for dynamic template classes
├── learning-objects/
│   ├── content/
│   ├── layout/
│   └── structure/
├── navigators/
├── time/
└── utils/
  
```





1. Reader App (apps/reader)

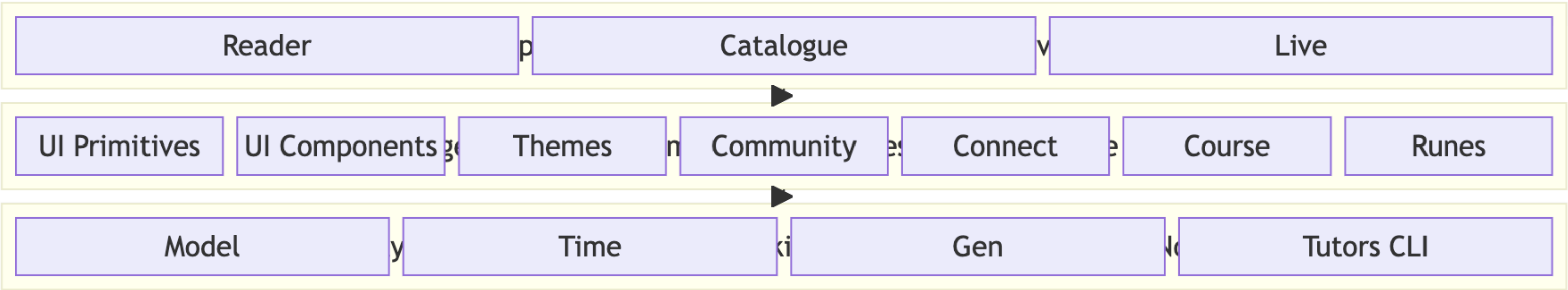
Purpose: Main course reader application

Route Groups:

- (auth)/ - Authentication routes
- (course-reader)/ - Course content routes
- (home)/ - Landing page and course list

Key Routes:

- / - Home/landing page
- /course/[courseid] - Course home
- /topic/[courseid]/[...loid] - Topic page
- /lab/[courseid]/[...loid] - Lab viewer
- /talk/[courseid]/[...loid] - Talk viewer
- /video/[courseid]/[...loid] - Video player
- /note/[courseid]/[...loid] - PDF viewer
- /notebook/[courseid]/[...loid] - Notebook viewer
- /time/[courseid] - Analytics dashboard
- /search/[courseid] - Course search

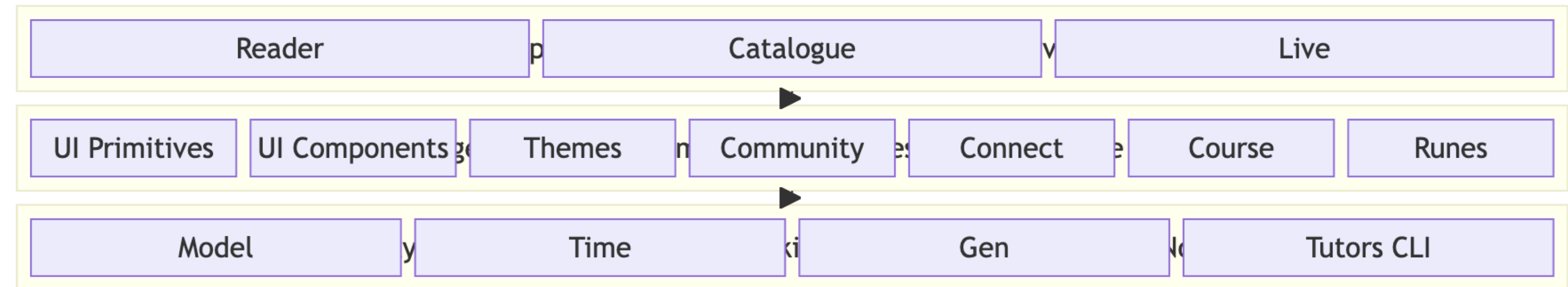


2. Catalogue App (apps/catalogue)

Purpose: Course catalog and discovery

Features:

- Course listing with statistics
- Visit tracking
- Minimal UI for browsing



3. Live App (apps/live)

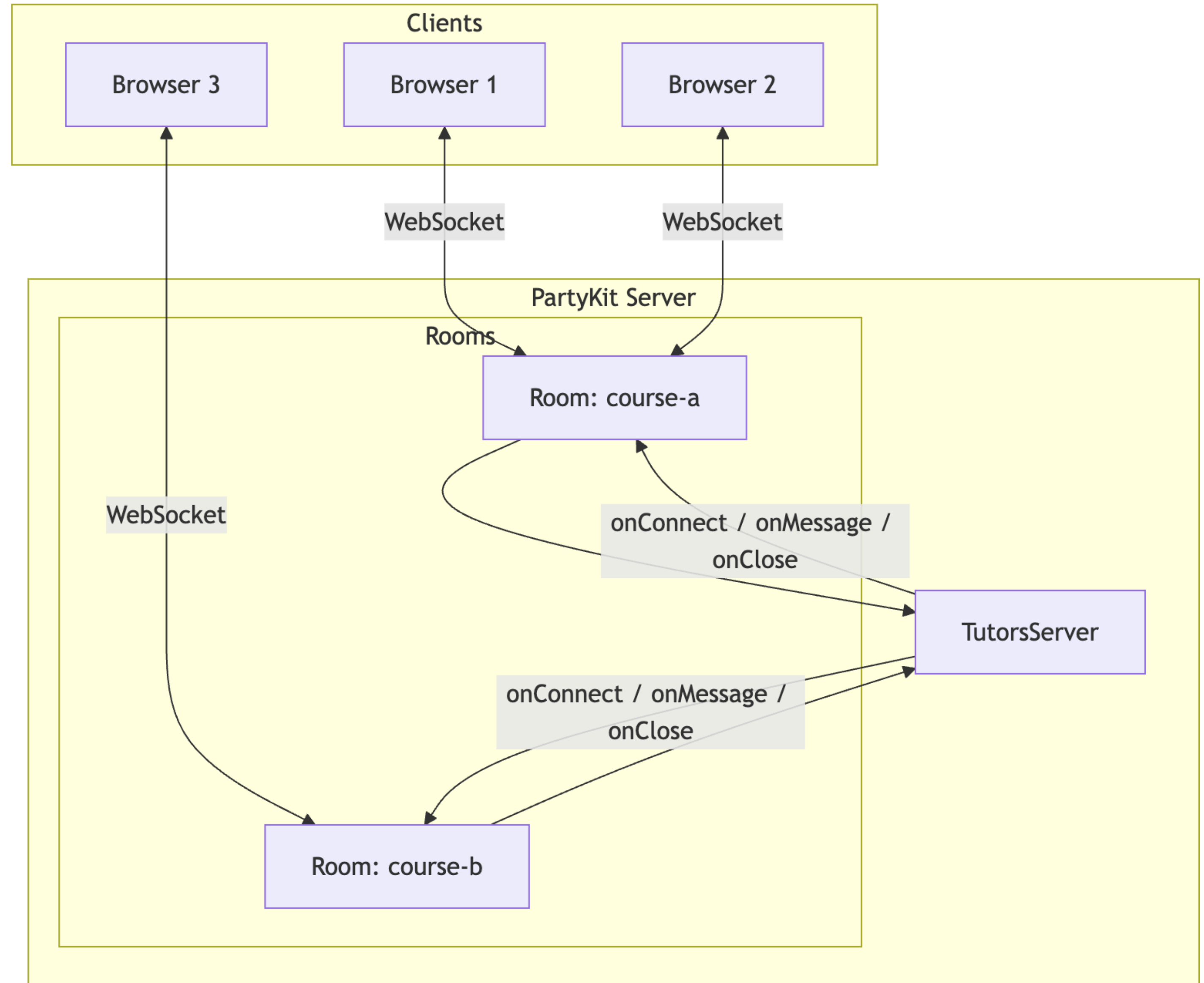
Purpose: Real-time student presence tracking

Features:

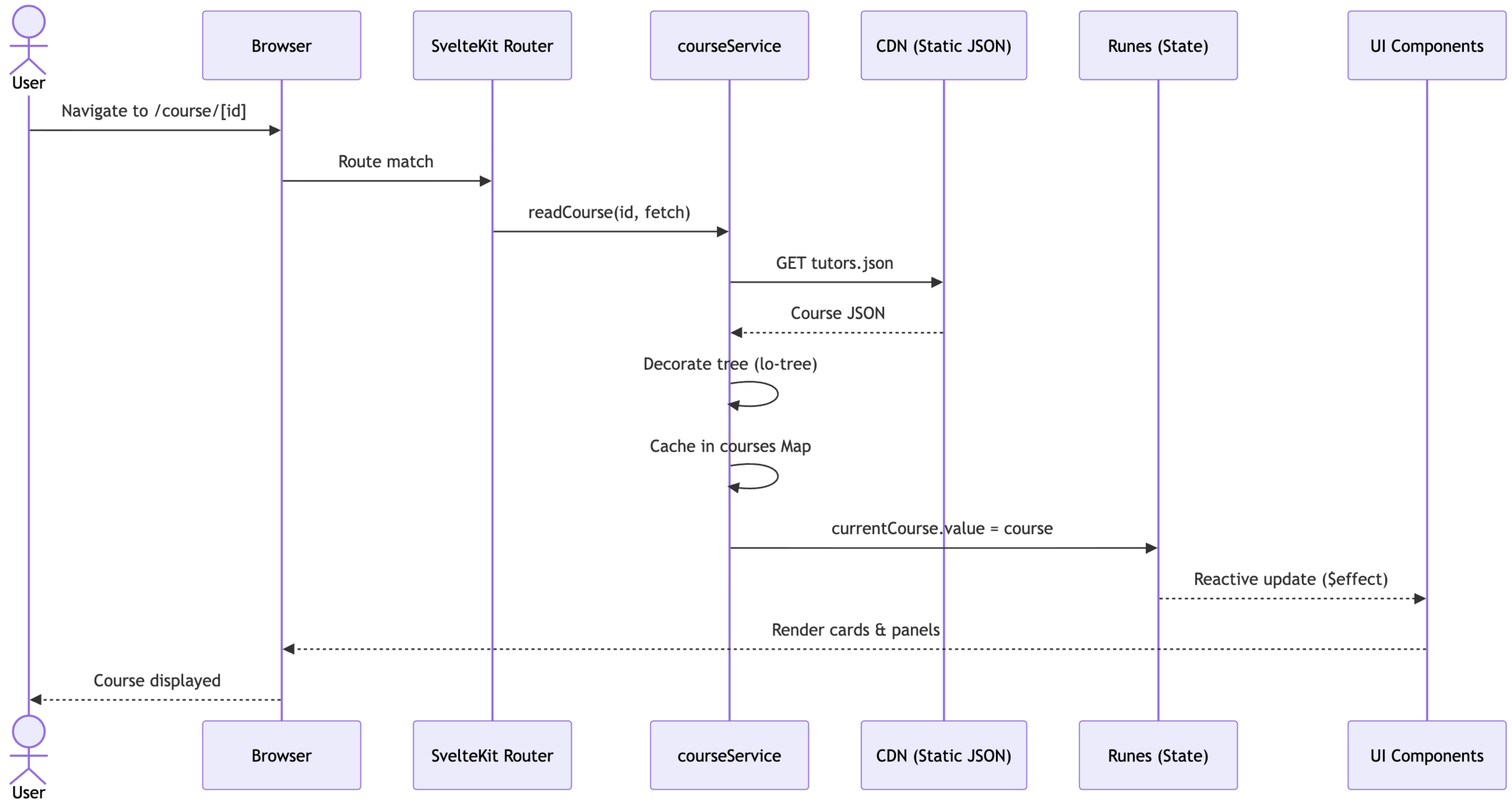
- Live student count per course
- Current page views
- Real-time updates via PartyKit

PartyKit Service (services/party)

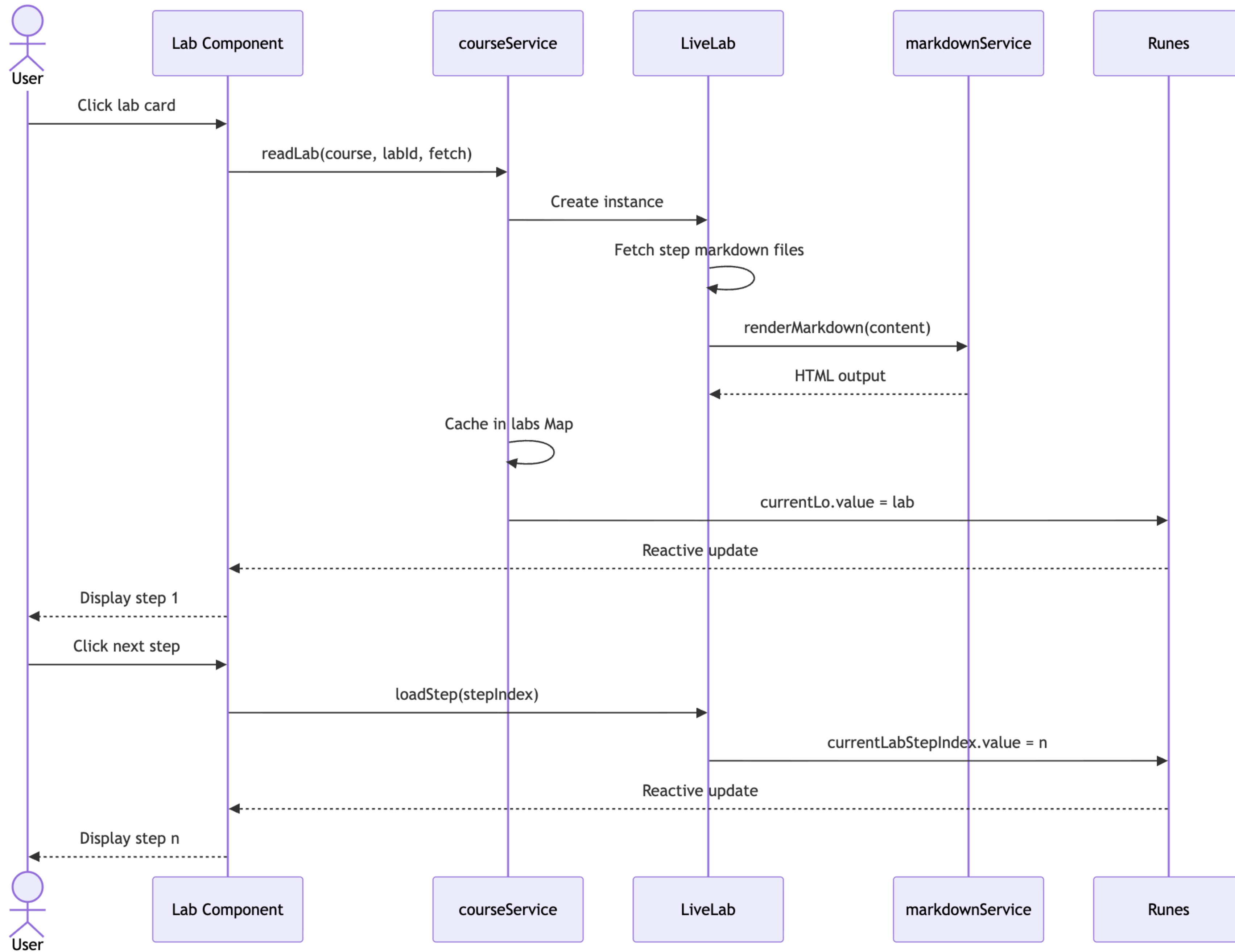
Purpose: Real-time WebSocket server for live presence

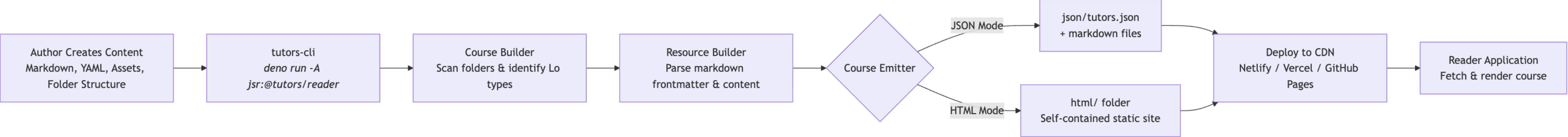


Course Loading Sequence



Lab Viewing Sequence





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